Parameter Tuning Research Summary in ERBOT

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Here I summarize the tuning parameters and research progress in the R package `erbot`, based on the previous codebase and the newly added tuning-enabled version. This summary is tightly aligned with the actual architecture and methods implemented in a new R package `ERBOT`, and highlights the research motivations, methods, benefits, and contributions compared with existing ER systems.

1 Introduction: Why ERBOT Requires a Rich Parameter Tuning Framework

Entity Resolution and Deduplication pipelines depend critically on feature extraction, blocking strategies, similarity construction, and clustering. Prior ER systems (e.g., Dedoop, Febrl, JedAI) expose several parameters but provide limited cross-method coordination and minimal automatic tuning.

`ERBOT` integrates feature-space clustering, graph-based community detection, classical deduplication, Graph Coloring (GCMER), and multifield similarity fusion. The new version enhances these components with a unified, systematic tuning mechanism, making ERBOT one of the most parameter-transparent and empirically tunable ER research platforms.

2 What ERBOT Looked Like Before Tuning (Previous Code)

The earlier version of ERBOT had:

- fixed or user-specified k for KMeans, HC, and PAM;
- fixed `knn_k` and `louvain_min_sim` for graph clustering;
- Sorted-Neighborhood + Edit Distance with fixed window/threshold settings;
- no threshold sweep for Graph Coloring;
- no tuning curves, no evaluation-driven parameter selection;

- no unified report summarizing tuning results.

Thus, methods were functional but parameters remained static and not research-friendly.

3 Major Parameter Groups in ERBOT

The major parameter groups include:

Feature Extraction

- `svd_dim` (dimension of TF–IDF + SVD)
- vocabulary size (implicit via `text2vec`)

This controls semantic richness vs. overfitting.

Clustering (Vector-Based)

- k for KMeans, HC, PAM
- linkage method (Ward.D2)

ERBOT supports silhouette-based tuning and ARI-based tuning.

Graph-Based Methods

- `knn_k`: number of neighbors
- `min_sim`: pruning threshold
- `cos_thresh`: embedding-based pruning

Edit-Distance + Sorted Neighborhood

- `sn_window`
- `sn_method` (JW, LV)
- `sn_thresh`
- `mst_cut_ratio`
- `mst_k`

Graph Coloring (GCMER)

- GC method (LF, SL, RLF)
- string distance method
- threshold sweep vector
- tuning metric (ARI, MI, silhouette)

Multi-Field Similarity Fusion

- field weights w
- numeric decay parameter τ
- similarity methods (JW, LV, Jaccard, categorical match)
- top-k similarity retention

4 What the New ERBOT Provides (With Full Tuning System)

The tuning-enhanced ERBOT introduces:

- automatic k selection via silhouette or ARI;
- threshold sweeps for graph coloring;
- pruning tuning for MST and similarity graphs;
- τ tuning for numeric/year similarity fields;
- side-by-side comparison of all clustering/deduplication methods;
- fully integrated PDF reporting with tuning curves and parameter tables.

5 Unified Auto-Tuning Workflow

1. Field selection and text normalization.
2. Feature extraction (TF–IDF + SVD; optional embeddings).
3. Execute KMeans, HC, PAM, SN+Edit, Louvain-kNN, embedding-kNN, GC.
4. Tune parameters via silhouette or external truth (ARI).
5. Evaluate using GCMER metrics.
6. Produce complete PDF report with curves, parameter tables, and performance comparisons.

6 Comparison with Existing Research and Tools

System	Auto-Tuning	Multi-Method	Numeric τ	SN+MST
Dedoop	No	Limited	No	No
Febrl	No	Limited	Limited	No
JedAI	Weak	Yes	No	No
GCMER	No	GC only	No	No
ERBOT	**Yes**	**Yes**	**Yes**	**Yes**

Summary: ERBOT provides the first R-based, fully tunable ER pipeline that systematically optimizes feature extraction, similarity fusion, classical deduplication, graph-based clustering, and Graph Coloring—supported by tuning curves, metrics, and a unified PDF reporting system. This positions ERBOT as a research-grade, multi-method ER framework that surpasses existing systems in tunability and transparency.